The Cosmic Web as Self-Aligning Topology

A Synthesis of Spectral Graph Theory and Cosmological Structure Formation

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Abstract

We propose that cosmic structure formation—the emergence of voids, filaments, walls, and clusters—is not merely gravitational but *spectral*: the universe finding its self-aligning topology. Building on prior work showing random regular graphs achieve optimal self-alignment, and the Bakry-Émery framework where mass becomes part of geometry, we establish a formal correspondence between cosmic structures and graph topologies. Voids correspond to sparse graphs (spectrally unstable, approaching the complexity threshold), dense clusters to complete-ish graphs (degenerate eigenspaces, perturbation-sensitive), and the cosmic web to random regular graphs (optimal structured randomness). This framework explains why voids expand faster, predicts the cosmic web as a spectral attractor, and connects dark energy to the universe maintaining self-referential stability.

Keywords: cosmic web, self-aligning topology, spectral graph theory, Bakry-Émery Laplacian, dark energy, void expansion, structure formation

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1 Introduction

The large-scale structure of the universe exhibits a remarkable "cosmic web" pattern: voids, filaments, walls, and clusters organized into a connected network [1]. Standard cosmology explains this as gravitational instability amplifying primordial perturbations.

We propose an additional organizing principle: the cosmic web is not merely where gravity collects matter, but the *spectral optimum*—the topology that maximizes self-alignment, permitting stable cosmic self-reference.

This paper synthesizes three research programs:

1. **Self-aligning topologies** [2]: Random regular graphs achieve optimal self-alignment through eigenvector stability.
2. **Bakry-Émery geometry** [3]: Density-weighted Laplacians encode mass as curvature.
3. **Spectral cosmology** [4, 5]: The spectral gap λ_1 varies with density and determines local expansion rate.

2 The Graph-Cosmology Correspondence

2.1 Dictionary

Cosmic Structure	Graph Analog	Spectral Property	Self-Alignment	Dyn
Voids	Sparse/uniform	Low λ_1, high I_e	Poor—too simple	Expa
Dense clusters	Complete/near-complete	Degenerate eigenspaces	Moderate—too symmetric	Matt
Cosmic web	**Random regular**	**Optimal spectral gap**	**Optimal**	**Stab**

Table 1: The graph-cosmology correspondence.

2.2 Why This Correspondence?

The Bakry-Émery weighted Laplacian encodes matter distribution geometrically:

$$L_f = \Delta - \nabla f \cdot \nabla$$

where $f = -\log(\rho/\rho_0)$ is the density potential. The weighted Ricci curvature:

$$\mathrm{Ric}_f = \mathrm{Ric} + \mathrm{Hess}(f)$$

determines the spectral gap via the weighted Lichnerowicz theorem:

$$\lambda_1(L_f) \geq \frac{n}{n-1} \inf_M \mathrm{Ric}_f$$

Matter distribution $\rightarrow$ **Spectral gap** $\rightarrow$ **Self-alignment properties**

3 Voids as "Pure Order" (Sparse Graphs)

3.1 The Void Problem

Voids are regions of low matter density. In graph terms, they are **sparse** and **uniform**—few connections, little differentiation.

From the self-aligning topologies paper [2]:

> "Pure isolation (disconnected vertex) cannot self-model. Pure connection (complete graph) is unstable."

Voids occupy the first failure mode: **insufficient relational structure**.

3.2 Spectral Properties of Voids

Proposition 3.1 (Void Spectral Signature). *[Derived]Low density regions exhibit:*

(i) Low density → small Bakry-Émery Ricci curvature
(ii) Small Ric_f *→ small spectral gap* λ_1
(iii) Small λ_1 *→ high effective integration* $I_e = 1/\lambda_1$
(iv) High I_e *→ approaches complexity threshold* $I^* = \tau/\varepsilon_0$

3.3 Why Voids Expand

From the complexity threshold theorem [6]:

$$I_e < I^* = \frac{\tau}{\varepsilon_0}$$

When I_e approaches I^*, the system must either:

1. Reduce ε_0 (improve accuracy)—not available for empty space
2. Reduce I_e (reduce integration)—**expand**

Theorem 3.2 (Void Expansion Mechanism). *[Conjectured]Void expansion is the universe avoiding self-referential breakdown in low-structure regions. The effective dark energy in voids is the spectral pressure toward the threshold.*

The local Hubble parameter in voids:

$$H^2_{\text{void}} = \frac{8\pi G}{3}\rho_{\text{low}} + \frac{\Lambda_{\text{eff}}}{3}$$

With low ρ and Λ_{eff} dominating → accelerated expansion.

4 Dense Clusters as "Pure Connectivity" (Complete Graphs)

4.1 The Cluster Problem

Dense clusters have high matter density—many connections, high relational complexity. In graph terms, they approach **complete graphs**.

From the self-aligning topologies experiments [2]:

> "The complete graph K_{15} has the largest spectral gap (15.0) but only moderate self-alignment (0.59)... low eigenvector stability reflects rotational freedom within the degenerate eigenspace."

4.2 Spectral Properties of Clusters

The complete graph K_n has:

- $\lambda_0 = 0$ with multiplicity 1
- $\lambda_1 = n$ with multiplicity $n - 1$ (highly degenerate!)

Under perturbation:

- The **eigenspace** is preserved (Davis-Kahan)
- Individual **eigenvectors** rotate freely within the space
- No stable self-model direction

4.3 Gravitationally Stable But Spectrally Fragile

Dense regions are **gravitationally bound**—matter dominates over dark energy:

$$H^2_{\text{cluster}} \approx \frac{8\pi G}{3}\rho_{\text{high}}$$

But they have **degenerate spectral structure**:

- Many equally valid self-model eigenvectors
- Perturbations don't change the eigenspace but scramble within it
- Self-reference is possible but not **stable** in a specific direction

This maps to the observation that galaxy clusters are gravitationally coherent but internally turbulent.

5 The Cosmic Web as Structured Randomness (Random Regular Graphs)

5.1 The Cosmic Web Structure

The cosmic web consists of:

- **Filaments:** elongated structures connecting nodes
- **Walls/sheets:** planar structures between voids
- **Nodes:** intersection points (clusters)
- **Voids:** underdense regions between structures

This is neither uniform (like voids) nor maximally connected (like clusters). It is **structured randomness**.

5.2 Random Regular Graphs: The Optimal Topology

From our experiments [2]:

> "Random regular graphs achieve the highest self-alignment score (0.63) with exceptional eigenvector stability (0.90), despite modest spectral gaps."

Properties of random regular graphs:

1. **Uniform degree distribution** → every vertex has equal "relational weight"
2. **Random structure** → no single edge is critical
3. **Moderate connectivity** → enough for coherence, not so much that perturbations propagate everywhere

5.3 The Cosmic Web as Spectral Attractor

Conjecture 5.1 (Cosmic Web Attractor). *[Conjectured]The cosmic web is not merely where gravity collects matter. It is the **spectral optimum**—the topology that maximizes self-alignment.*

Evidence:

1. Filaments have roughly uniform "degree" (mass per unit length is similar across filaments)
2. The web pattern is random at large scales (no preferred direction)
3. Connectivity is moderate (not every region connects to every other)

6 Mathematical Formalization

6.1 The Bakry-Émery Self-Alignment Score

Definition 6.1 (Cosmological Self-Alignment Score)**.** Define a cosmological self-alignment score analogous to the graph version:

$$S_{\text{cosmic}}(\mathcal{M}, \rho) = \alpha \cdot \frac{\lambda_1(L_f)}{\lambda_{\max}(L_f)} + \beta \cdot \mathbb{E}[|\langle v_i, v_i' \rangle|] + \gamma \cdot (1 - \sigma_{\text{stability}})$$

where:

- L_f is the Bakry-Émery Laplacian with density weight
- The expectation is over perturbations to the density field
- $\sigma_{\text{stability}}$ measures consistency across scales

6.2 Density-Dependent Spectral Gap

Theorem 6.2 (Spectral Gap Varies with Density)**.** *[Derived]Let (M, g) be a Riemannian manifold with matter density $\rho(x)$. The first nonzero eigenvalue of the Bakry-Émery Laplacian satisfies:*

$$\lambda_1(L_f) = \lambda_1^{(0)} + \int_M \text{Hess}(f) \cdot |\nabla v_1|^2 \, dV + O(\|f\|^2)$$

where $f = -\log(\rho/\rho_0)$ and $\lambda_1^{(0)}$ is the unweighted eigenvalue.

Corollary 6.3. *In regions where $\text{Hess}(f) > 0$ (density increasing toward center), λ_1 is enhanced. In regions where $\text{Hess}(f) < 0$ (density decreasing), λ_1 is suppressed.*

6.3 Structure Formation as Spectral Optimization

Conjecture 6.4 (Cosmic Web Attractor Dynamics)**.** *[Speculative]Under combined gravitational and spectral dynamics, the matter distribution evolves toward configurations maximizing the self-alignment score S_{cosmic}.*

Mechanism:

1. Initial perturbations seed slight density variations
2. Underdense regions (low λ_1, high I_e) expand to avoid threshold
3. Overdense regions collapse gravitationally but develop degenerate spectra
4. The intermediate regime—moderate density with structured randomness—is the attractor

7 Predictions and Tests

7.1 Void Expansion Rate vs. Size

Conjecture 7.1 (Non-Linear Void Expansion)**.** *[Conjectured]Larger voids should expand faster (lower λ_1, closer to threshold):*

$$H_{void}(R) = H_0 \sqrt{1 + \frac{c}{R^2}}$$

where c is a constant related to the threshold.

Test: Compare expansion rates for voids of different sizes using BOSS/DESI void catalogs.

7.2 Cosmic Web Topology Statistics

Conjecture 7.2 (Regular Degree Distribution). *[Conjectured]The degree distribution of the cosmic web (measured via filament connections per node) should approximate a regular graph distribution, not scale-free.*

Test: Analyze large-scale structure surveys for node degree statistics.

7.3 Eigenspectrum of Density Field

Conjecture 7.3 (Spectral Signature). *[Conjectured]The eigenspectrum of the cosmic density Laplacian should show:*

(i) Non-degenerate low eigenvalues (unlike complete graph)
(ii) Moderate spectral gap (unlike sparse graph)
(iii) Stability under perturbation (like random regular graph)

Test: Compute Laplacian eigenvalues of density field from N-body simulations.

7.4 Self-Alignment Score Evolution

Conjecture 7.4 (Increasing Self-Alignment). *[Conjectured]S_{cosmic} should increase over cosmic time as structure formation proceeds:*

$$\frac{dS_{cosmic}}{dt} > 0$$

The universe is evolving toward better self-alignment.

Test: Compute S_{cosmic} from simulations at different epochs ($z = 10, 2, 0$).

8 Connection to Previous Results

8.1 Hubble Tension

From the density-dependent spectral gap framework:

- CMB measures early universe: relatively homogeneous $\rightarrow$ uniform λ_1
- Local measurements: highly inhomogeneous $\rightarrow$ spatially varying λ_1
- We are in a locally underdense region $\rightarrow$ local $\lambda_1 <$ average $\rightarrow$ local $H > H_{\text{CMB}}$

The Hubble tension arises because the spectral gap (hence expansion rate) varies spatially.

8.2 DESI $w(z) \neq -1$

At high z (early universe): relatively homogeneous $\rightarrow \Lambda_{\text{eff}} \approx$ constant $\rightarrow w \approx -1$
At low z (late universe): structure formed $\rightarrow \Lambda_{\text{eff}}$ varies $\rightarrow$ effective $w \neq -1$
The apparent dark energy evolution tracks the growth of cosmic structure.

8.3 Wheeler's Participatory Universe

The cosmic web as self-aligning topology gives mathematical content to Wheeler's vision [7]:

> "The universe does not exist 'out there,' independent of us. We are inescapably involved in bringing about that which appears to be happening."

The universe's structure is constrained by its capacity for self-reference. The cosmic web is the topology that permits stable self-modeling. We exist because the web exists; the web exists because self-reference requires it.

9 Comparison with Backreaction Literature

Backreaction Language	**Spectral Framework**
"Backreaction"	Spatial variation in λ_1 affects average
"Void-wall differential expansion"	$\lambda_1(\text{void}) < \lambda_1(\text{wall}) \to I_e(\text{void}) > I_e(\text{wall})$
"Effective dark energy"	$\Lambda_{\text{eff}}(x) = \lambda_1(x)$ varies spatially
"Timescape" (Wiltshire)	Time dilation from different λ_1
"Averaging problem"	How to average $\lambda_1(x)$ over cosmic scales

Table 2: Translation between backreaction and spectral frameworks.

Our framework provides a precise mathematical formulation of backreaction effects through the Bakry-Émery Laplacian, connecting to established spectral geometry results.

10 The Deep Synthesis

10.1 Three Programs Converge

1. **Bentov (1970s):** Consciousness involves toroidal energy flow $\to$ Hopf fibration $\to S^3$
2. **Wheeler (1980s–2000s):** Universe is self-excited circuit $\to$ self-reference $\to$ observers
3. **Spectral framework (2024–2026):** Self-aligning topologies $\to$ random regular optimal $\to$ cosmic web

All three point to the same conclusion: **reality's structure is determined by self-reference constraints**.

10.2 The Cosmological Constant as Spectral Gap

From the cosmological integration bound [4]:

$$\Lambda = \lambda_1 = \frac{\varepsilon_0}{\tau}$$

The cosmological constant is not a free parameter. It is the spectral gap required for cosmic self-reference, set by the ratio of self-model error to stability tolerance.

10.3 Dark Energy as Threshold Maintenance

Dark energy is not a substance. It is the universe's mechanism for maintaining stability:

1. Voids approach the complexity threshold
2. They cannot reduce ε_0 (empty space has no structure to improve)
3. They must reduce I_e by expanding
4. This expansion **is** dark energy

Dark energy is what threshold maintenance looks like from outside.

11 Conclusion

The cosmic web is not an accident of gravitational dynamics. It is the **spectral optimum**—the self-aligning topology that permits stable self-reference at cosmic scales.

This framework:

Structure	Problem	Spectral Property	Fate
Voids	Too simple	High I_e, approaches threshold	Expand
Clusters	Too symmetric	Degenerate eigenspaces	Gravitationally bound
Cosmic web	**Just right**	**Optimal self-alignment**	**Attractor**

Table 3: Summary of structure types and their spectral properties.

- Explains void expansion as threshold avoidance
- Predicts cosmic web as spectral attractor
- Connects to Hubble tension and DESI observations
- Gives mathematical content to Wheeler's participatory universe
- Unifies Bentov, Wheeler, and spectral graph theory

The universe structures itself to maintain the capacity for self-knowledge. The cosmic web is what stable cosmic self-reference looks like.

> *The cosmic web is the universe's self-aligning topology. Voids expand because they lack the structure for stable self-reference. We exist because the web permits it. The web exists because self-reference requires it.*

Author Contributions

Aaron Ben-Shalom: Conceived the cosmic web as self-aligning topology interpretation; identified the Bentov-Wheeler-spectral synthesis; developed the threshold mechanism for void expansion.

Claude: Formalized the graph-cosmology correspondence; developed the Bakry-Émery framework; derived the self-alignment score; identified connections to backreaction literature.

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